A prescribed OH perturbation in Laughner et al. (2021)

Reference. Laughner et al., *PNAS* 118, e2109481118, doi:10.1073/pnas.2109481118. The main text, 19-page SI Appendix, team list, and both supplementary workbooks (SI Tables S2 and S4) were read.

Method. This synthesis combines pandemic activity indicators, observations, and chemistry experiments. Its methane analysis uses TCCON columns at Park Falls and Lauder. Flag-zero data are averaged over 15 days, deseasonalized, and compared with a robust linear 2016–2019 trend. Growth rates use differences between three-month means multiplied by four. Figure 6b averages monthly fractional deviations from each site's trend; it is not a global mass-balance growth anomaly. A two-hemisphere methane–OH box model includes emissions, oxidation, and interhemispheric transport. Methane emissions follow a stabilized scenario. The plotted perturbation compares runs with the same emissions, one having OH reduced by 3% in 2020.

Findings and uncertainty. Methane columns rise about 0.3% above their extrapolated trends in 2020, similar to the OH scenario. The authors interpret this as support for a contribution from COVID-related NO_x reductions and lower OH, while allowing emission changes. The 2–4% OH range comes from temporal variation in a chemistry-assimilation experiment: 2% in March and 4% in May. It is a proxy for uncertainty, not propagated posterior uncertainty in an annual global OH estimate. The two 2021 lines are extrapolations: one holds the January 2021 concentration difference fixed, and one extends its final-six-month 2020 slope. They were not calculated with the box model.

Interpretation for a 2019 baseline. A uniform fractional OH reduction r gives the fixed-burden forcing

$$F_{\mathrm{OH},2020} = r\, m_{2019}/\tau_{\mathrm{OH},2019}, \tag{1}$$

in which m is methane burden and τ_{OH} is the OH-only lifetime. We evaluate this conditional scenario using a NOAA burden proxy and an 11.1-yr benchmark, with 9–12-yr sensitivity. It is not a native flux reconstruction or reproduction of the two-site observational anomaly. The imposed 2020 forcing is zero again in 2021–2022; the accumulated concentration response need not be zero.

Numerical data. Dataset S1 gives CO_2, CH_4, and NO_x inventory series. Its CH_4 column ends in 2017; 2018–2020 are missing. The equivalent-emission-year illustration uses extrapolation and an assumed reduction, not an inferred 2020 global methane flux. Dataset S2 gives monthly ozone-production efficiencies for 20 cities, not methane sink fluxes. Both workbooks are retained as numerical transcriptions. No 2019–2022 methane budget is filled from their missing entries.